

MODEL QUESTION PAPER

**TED (10) 5020. FIFTH SEMESTER DIPLOMA EXAM IN AUTOMOBILE ENGG
REV 2010**

SUB: FUELS AND COMBUSTION

(Max Mark: 100.

Time: 3 hrs)

PART A

Answer the following in one or two sentences. Each question carries 2 marks.

1. Name two different alcohols used as SI engine fuels
2. Write the chemical formula of petrol
3. Define swirl
4. State the difference in the working of a super charger and a turbo charger
5. Write the function of condenser in an air conditioning system? (5x2 = 10)

PART B

Answer any six of the following question. Each question carries 5 marks.

1. Discuss the advantages of LPG as an IC engine fuel
2. Discuss the requirements of fuel additives
3. Discuss the normal ranges of combustion for SI and CI engines
4. Compare the CI and SI engines for the following factors: a) the initial cost b) the maintenance cost.
5. CI engines are more difficult to start. Give reasons
6. Diesel engines are more suitable for supercharging. Give reasons
7. Write the function of condenser in an air conditioning system. (6x5 = 30)

PART C

Answer any one full question from each unit. Each full question carries a total of 15 marks.

UNIT 1

- a) Discuss the basic properties required for a good IC engine fuel (8)
- b) Discuss Octane number. How is it found? (7)

OR

- a) Discuss the use of CNG and LPG as IC engine fuels (8)
- b) Can petrol be used as a Diesel engine fuel? Discuss. (7)

UNIT 2

- a) Define combustion, discuss the general conditions necessary for combustion (8)
- b) Define ignition lag and discuss the various factors affecting ignition lag (7)

OR

- a) Discuss the effects the following engine variables on flame propagation (8)
- b) Discuss the main factors which affect the tendency to detonate in SI engines (7)

UNIT 3

- a) Discuss the main advantages and applications of Diesel engines (8)
- b) Discuss CI engine combustion (7)

OR

- a) Discuss Diesel knock (8)
- b) Discuss the air-fuel ratios used in CI engines (7)

UNIT 4

- a) Discuss the supercharging of CI engines (8)
- b) Discuss the function and operation of expansion valve in an air conditioning system. (7)

OR

- a) Write down the limitations of turbocharging
- b) Discuss the desirable properties of a refrigerant